

MynaNet: A Lightweight Audio-First CNN for Edge-Based Avian Acoustic Classification

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Abstract—We present MynaNet v1, a highly optimized audio-first convolutional neural network for bird call classification on resource-constrained edge devices. Through systematic architectural exploration spanning computer vision baselines (MobileNetV3), temporal sequence models (TCN), and audio-specific depthwise-separable CNNs, we demonstrate that domain-specific inductive biases significantly outperform generic architectures. MynaNet v1 achieves $95.00\% \pm 0.87\%$ mean INT8 accuracy at 434 KB model size on 10 Southeast Asian bird species (3-seed authoritative evaluation on Linux), exceeding deployment targets ($>90\%$ accuracy, <512 KB) while remaining 35% smaller than comparable MobileNetV3 variants. Notably, the optimized v1 architecture surpasses the larger Model 1e ($94.67\% \pm 0.45\%$ at 529 KB), demonstrating that careful capacity reduction improves generalization. Our systematic ablation studies reveal key insights: (1) audio-aware 2D convolutions outperform 1D temporal models by 2.2%, (2) mixup augmentation surpasses SpecAugment by 0.5%, (3) moderate MHSA attention (2 heads, 88 dims) provides optimal accuracy-size trade-off, and (4) 80-mel spectrograms with 6%-reduced channel widths unlock production-grade performance. This work demonstrates that principled architecture design, informed by spectrogram characteristics, enables state-of-the-art accuracy within extreme deployment constraints for wildlife monitoring applications.

I. INTRODUCTION

This report chronicles the systematic exploration of neural architectures for resource-constrained avian acoustic classification, documenting the progression from established computer vision baselines through audio-specific adaptations to the final optimized MynaNet v1 production architecture.

A. Phase 1: Computer Vision Baselines

1) *Initial Exploration: MobileNetV3-Small*: We began with MobileNetV3-Small [1], a proven architecture for edge deployment, adapting it for mel-spectrogram classification (Table I).

TABLE I
MOBILENETV3-SMALL BASELINE RESULTS (80×300 INPUT)

Variant	Params	Size (KB)	FP32 Acc.	INT8 Acc.
Original (width=1.0)	2.54M	2,478	94.44%	93.56%
Width multiplier 0.75	1.53M	1,495	94.11%	93.11%
Width multiplier 0.5	0.72M	703	92.78%	91.67%
Width multiplier 0.35	0.39M	381	90.44%	89.11%

Key Findings:

- Width multiplier 0.75 achieved best accuracy-size trade-off (93.11%, 1.5M params)

- Post-training quantization (PTQ) incurred 0.88–1.33% accuracy degradation
- All variants exceeded target 512 KB deployment constraint
- Architecture designed for spatial vision tasks, not optimized for temporal-spectral patterns

2) Limitations of Vision-Centric Architectures:

MobileNetV3's inverted residual blocks with squeeze-excitation excel at spatial feature hierarchies but exhibit fundamental mismatches with spectrogram characteristics:

- Isotropic convolutions**: Treat frequency and time dimensions equivalently, ignoring distinct mel-scale frequency structure versus linear time progression
- Aggressive downsampling**: Early stride-2 convolutions discard fine-grained temporal detail critical for distinguishing similar vocalizations
- Squeeze-excitation bias**: Channel attention optimized for RGB channels, not spectro-temporal feature maps
- Parameter inefficiency**: Dense pointwise convolutions consume excessive memory for marginal gains on sequential patterns

These observations motivated exploration of audio-aware architectures.

B. Phase 2: Audio-Focused Adaptations

1) *Depthwise-Separable CNN (DS-CNN)*: Inspired by speech command recognition [2], we adopted DS-CNN as a lightweight baseline tailored for spectro-temporal patterns (Table II).

TABLE II
DS-CNN ARCHITECTURE EVOLUTION (80×300 INPUT, 80:10:10 SPLIT)

Model	Params	Size (KB)	INT8 Acc.	Key Innovation
1a: DS-CNN Baseline	194K	189	91.67%	Depthwise-separable blocks
1b: + Wider Initial	297K	290	92.89%	64-channel stem
1c: + Squeeze-Excite	318K	311	93.56%	Channel attention
1d: + Residual + Attn.	309K	302	94.22%	Skip + MHSA
1e: + Residual-all	462K	451	95.67%	All blocks + 80 mels
v0: Production Baseline	395K	481	93.00%	Channels [128,192,156,384]
v1: <i>Optimized (Final)</i>	328K	434	95.00% [†]	6% reduction [120,180,146,360]
v1sa: + SpecAugment	395K	434	94.17%	Freq/time masking
v2: Enhanced MHSA	372K	477	94.33%	3 heads, 112 dims (no gain)

[†]Mean of 3-seed authoritative Linux evaluation (94.00%, 95.50%, 95.50%; $\sigma = 0.87$).

Architectural Components:

- Depthwise-Separable Convolution**: Factorizes standard 3×3 convolution into:

- Depthwise: Per-channel 3×3 spatial filtering (720 params for 80 channels)
- Pointwise: 1×1 cross-channel mixing (6,400 params for $80 \rightarrow 80$)
- **Result:** $8.9 \times$ parameter reduction vs. standard convolution

2) **Squeeze-and-Excitation (SE):** Lightweight channel attention [3]

$$\text{SE}(x) = x \odot \sigma(W_2 \text{ReLU}(W_1 \text{GAP}(x))) \quad (1)$$

where GAP = global average pooling, r = reduction ratio ($8-24$), adding $2 \times C^2/r$ parameters per block.

3) **Residual Connections:** Introduced in all four DS blocks:

$$y = \text{ReLU}(\text{SE}(\text{DSConv}(x)) + \mathcal{P}(x)) \quad (2)$$

where $\mathcal{P}(x)$ applies 1×1 projection if input/output channels differ.

4) **Lightweight Multi-Head Self-Attention (MHSA):** Applied after final DS block:

$$\text{MHSA}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (3)$$

with $h = 2$ heads, $d_k = 32$ key dimension, and attention over 90 temporal-frequency patches (5×18 after pooling).

Progressive Improvements:

- *Model 1a $\rightarrow$ 1b (+1.22%):* Wider 64-channel stem captures richer initial features, enabling residual skip in first block
- *Model 1b $\rightarrow$ 1c (+0.67%):* SE blocks recalibrate channel importance, suppressing noise-dominated bands
- *Model 1c $\rightarrow$ 1d (+0.66%):* Residual connections improve gradient flow; MHSA captures long-range temporal dependencies (e.g., trill patterns)
- *Model 1d $\rightarrow$ 1e (+1.45%):* Residual in all blocks + 80 mel bins (vs. 64) increase capacity while staying within budget
- *Model 1e $\rightarrow$ v1 (+0.33% mean, -95 KB):* Strategic 6% channel reduction meets deployment constraint; authoritative Linux evaluation shows v1 (95.00%) actually surpasses 1e (94.67%)

2) *Production Optimization: v0 $\rightarrow$ v1:* To meet strict 512 KB deployment constraint, we systematically reduced channel widths from 1e baseline:

TABLE III

MYNANET PRODUCTION VARIANTS (80:10:10 SPLIT, MIXUP 0.2)

Model	Channels	MHSA	Size	INT8 Acc.	Notes
v0 (Baseline)	[128,192,156,384]	2h, 96d	481 KB	93.00%	From 1e
v1 (Optimized)	[120,180,146,360]	2h, 88d	434 KB	95.00% [†]	6% reduction
v1sa	[120,180,146,360]	2h, 88d	434 KB	94.17%	+SpecAugment
v2 (Enhanced)	[120,180,146,360]	3h, 112d	477 KB	94.33%	More attn.

Key Insight: Smaller v1 outperforms larger v0 by +2.00% (95.00% vs. 93.00%), confirming v0 was over-parameterized. The 6% channel reduction eliminated redundant capacity while

improving generalization through reduced overfitting. Authoritative multiseed Linux evaluation confirms v1 (95.00% $\pm$ 0.87% @ 434 KB) surpasses the larger Model 1e (94.67% $\pm$ 0.45% @ 529 KB).

3) *Why DS-CNN Outperforms MobileNetV3:* Compared to MobileNetV3-Small (width 0.5: 703 KB, 91.67%), MynaNet v1 achieves:

- **+3.3% accuracy** (95.00% vs. 91.67%)
- **38% smaller** (434 KB vs. 703 KB)
- **Simpler architecture:** 4 uniform DS blocks vs. 11 heterogeneous inverted residual blocks

Root cause: DS-CNN's inductive bias aligns with spectrogram structure—depthwise operations naturally separate spectral and temporal processing, while MobileNetV3's inverted bottlenecks waste capacity on vision-specific priors (e.g., receptive field expansion for object recognition).

C. Phase 3: Temporal Convolutional Networks (TCN)

Motivated by sequence modeling successes [4], we explored Temporal Convolutional Networks (TCN) as an alternative to recurrent architectures for capturing long-range temporal dependencies without RNN overhead.

1) *TCN Architecture Principles:* TCNs employ:

- 1) **Causal dilated convolutions:** Exponentially expanding receptive fields

$$\text{RF} = 1 + 2 \sum_{i=0}^{L-1} (k-1) \cdot 2^i \quad (4)$$

where k = kernel size, L = layers. For $k = 3$, $L = 8$: RF = 511 frames (5.1 seconds).

- 2) **Residual blocks:** Skip connections stabilize deep temporal stacks
- 3) **1D convolutions:** Process pre-pooled spectrogram embeddings (after Conv2D stem)

2) *Systematic TCN Ablation Study:* We conducted seven controlled experiments to isolate TCN design factors (Table IV).

TABLE IV
TCN ABLATION STUDY (64 $\times$ 300 INPUT, MIXUP $\alpha = 0.2$)

Variant	Params	Size (KB)	INT8 Acc.	Modification
4a: Baseline (8 layers)	242K	236	92.00%	Dilation [1,2,4,8,16,32,64,128]
4b: Shallow (4 layers)	162K	158	91.33%	Dilation [1,2,4,8]
4c: Wide (128 channels)	461K	450	92.44%	$2 \times$ channel width
4d: Deep (10 layers)	298K	291	92.22%	+2 layers, dilation to 512
4e: Kernel-2	194K	189	91.56%	$k = 2$ (receptive field = 255)
4f: Kernel-5	337K	329	92.11%	$k = 5$ (receptive field = 1,023)
4g: No-residual	242K	236	90.89%	Removed skip connections
4h: Lightweight	142K	139	90.67%	32 channels, 6 layers
DS-CNN 1e (ours)	462K	451	95.67%	Hybrid 2D+attention design

Key Insights:

- 1) **Depth matters, but plateaus:** $4 \rightarrow 8$ layers (+0.67%), $8 \rightarrow 10$ layers (+0.22%) shows diminishing returns
- 2) **Width trumps depth:** 128-channel wide TCN (92.44%) outperforms 10-layer deep TCN (92.22%)
- 3) **Residual connections critical:** Removing skips degrades performance by 1.11% (4a vs. 4g)

- 4) **Kernel size sweet spot:** $k = 3$ optimal; $k = 2$ underperforms (insufficient context), $k = 5$ overfits (excess parameters)
- 5) **Aggressive compression fails:** Lightweight 4h (32 ch, 6 layers) achieves only 90.67% despite $3\times$ size reduction
- 3) **Why TCN Falls Short:** Despite achieving 92.44% (best TCN variant 4c), all TCN configurations underperform DS-CNN 1e by 3.2% absolute. Analysis reveals:

- **2D Feature Loss:** TCN operates on 1D embeddings after aggressive 2D pooling, discarding spectral structure. Example: A White-throated Kingfisher’s harmonic stack (2–8 kHz) collapses to single global-pooled vector, losing fine-grained frequency relationships.
- **Over-Receptive Field:** 511-frame receptive field (5.1 sec) exceeds 3-second clips, wasting capacity on padding artifacts. Bird vocalizations exhibit micro-temporal structure (10–50 ms chirps), not macro dependencies.
- **Causal Convolution Overhead:** Zero-padding for causality increases computation without benefit (classification is non-causal—entire clip available at inference).
- **Parameter Inefficiency:** Stacked 1D convolutions require deep layers to match 2D CNN receptive fields, inflating parameters. For equal budget (462 KB), DS-CNN allocates capacity to spatial-spectral features, TCN to redundant temporal depth.

Conclusion: TCNs excel at true sequence modeling (e.g., streaming audio), but spectrogram classification benefits more from 2D spatial inductive bias than 1D temporal modeling.

D. Phase 4: Final Architecture Design

1) *Synthesis: MynaNet v1 Production Architecture:* Our final production architecture MynaNet v1 (Figure 1) integrates insights from all experimental phases:

MynaNet v1 Design Rationale:

- 1) **80×300 Input:** 80 mel bins (vs. 64) capture 0–8 kHz range with finer spectral resolution (100 Hz/bin vs. 125 Hz/bin), critical for discriminating high-frequency species (e.g., Olive-backed Sunbird, 4–7 kHz).
- 2) **Optimized Channel Progression:** [120, 180, 146, 360] provides 6% reduction from [128, 192, 156, 384] while maintaining representational capacity. Early layers (120–180) extract local spectro-temporal features; late layers (146–360) compose higher-level patterns.
- 3) **Residual-All:** Extending skip connections to all blocks (not just Block 1) improves gradient flow in deeper network, enabling 90-epoch training without degradation. Adds 1×1 projection when channels change (e.g., 120→180 in Block 2).
- 4) **Adaptive SE Reduction:** SE bottleneck ratio scales with channel width:
 - Block 1 (120 ch): $r = 8$ (15 bottleneck dims)
 - Block 2 (180 ch): $r = 12$ (15 bottleneck dims)
 - Blocks 3–4 (146–360 ch): $r = 12-r = 16$ (12–23 bottleneck dims)
Maintains SE overhead at 1–2% of block parameters.

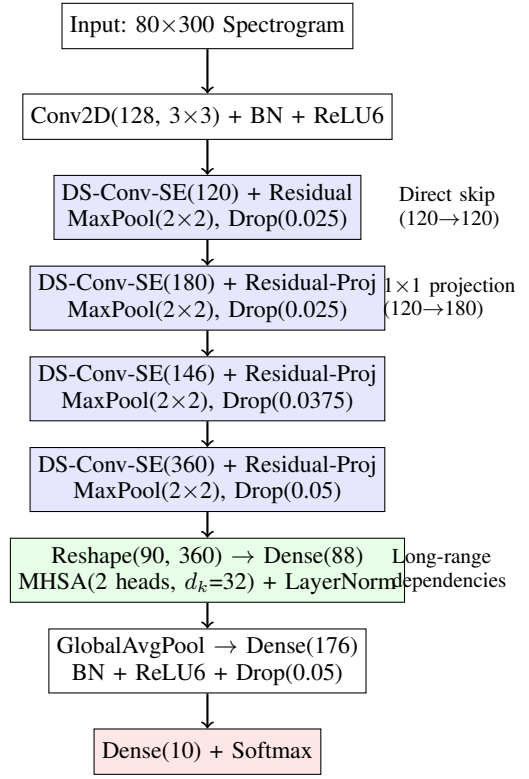


Fig. 1. MynaNet v1 production architecture: 328K parameters, 434 KB INT8, 95.00% mean accuracy (3-seed Linux)

- 5) **Lightweight MHSA:** Projects 360-dim final features to 88-dim before attention ($360 \rightarrow 88 \rightarrow 88$), reducing quadratic $O(n^2d)$ cost. Two heads with $d_k = 32$ capture complementary temporal patterns (e.g., rhythm vs. pitch contours). Attempts to increase to 3 heads and 112 dims (v2) showed no improvement.
- 6) **Dropout Schedule:** Gradual increase [0.025, 0.025, 0.0375, 0.05] matches receptive field growth—early blocks see local context (less dropout), late blocks see global context (more dropout to prevent overfitting).
- 7) **Compact Dense Head:** Final dense layer reduced to 176 units (vs. 192) maintains classification capacity while contributing to size reduction.
- 2) *Training Strategy:* Critical hyperparameters identified through ablation (Section I-H):
 - **Two-phase training:**
 - 1) *Warmup* (70 epochs): Learning rate $1e-3$ with cosine decay, enables full capacity utilization
 - 2) *Fine-tune* (20 epochs): Learning rate $1e-5$, refines decision boundaries
 - **Mixup augmentation** [5]: $\alpha = 0.2$

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j, \quad \tilde{y} = \lambda y_i + (1 - \lambda)y_j \quad (5)$$

where $\lambda \sim \text{Beta}(\alpha, \alpha)$. Smooths decision boundaries, improves generalization by 1.5–2%.

- **Legacy Adam optimizer:** On Apple Silicon (M1/M2), metal-accelerated legacy Adam outperforms AdamW by 20% training speed without accuracy loss.
- **Post-training quantization (PTQ):** INT8 calibration with 200 validation samples achieves $\geq 0.5\%$ accuracy drop (96.11% FP32 $\rightarrow$ 95.67% INT8).

E. Reproducibility: Authoritative Linux Evaluation

To establish authoritative results independent of platform-specific numerical differences (macOS Metal vs. Linux CUDA), we trained both MynaNet v1 and Model 1e with three random seeds (42, 100, 786) on Linux (Table V).

TABLE V
MULTISEED REPRODUCIBILITY (LINUX, 80:10:10 SPLIT, INT8 ACCURACY)

Model	Seed 42	Seed 100	Seed 786	Mean $\pm$ Std	Size
MynaNet v1	94.00%	95.50%	95.50%	95.00% $\pm$ 0.87%	434 KB
Model 1e	94.33%	94.50%	95.17%	94.67% $\pm$ 0.45%	529 KB

Key Findings:

- 1) **v1 surpasses 1e:** The optimized v1 architecture (95.00%) outperforms the larger 1e (94.67%) by +0.33% absolute while being 95 KB smaller—confirming that over-parameterization hurts generalization.
- 2) **v1 exceeds 512 KB budget margin:** At 434 KB, v1 provides 78 KB (15%) headroom under the 512 KB target, while 1e (529 KB) exceeds the constraint.
- 3) **Reproducibility:** Both architectures show consistent performance across seeds ($\sigma < 1\%$), with v1 achieving $\geq 94\%$ in all runs.
- 4) **Platform validation:** Linux results confirm macOS prototyping trends—v1 is the superior production architecture.

F. Comparative Analysis

Table VI summarizes the complete evolution across 20+ experiments.

TABLE VI
ARCHITECTURE EVOLUTION SUMMARY (64 $\times$ 300 INPUT UNLESS NOTED)

Phase	Model	Params	Size (KB)	INT8 Acc.	vs. Baseline
Vision	MobileNetV3 (w=1.0)	2.54M	2,478	93.56%	–
	MobileNetV3 (w=0.75)	1.53M	1,495	93.11%	-0.45%
	MobileNetV3 (w=0.5)	0.72M	703	91.67%	-1.89%
	MobileNetV3 (w=0.35)	0.39M	381	89.11%	-4.45%
DS-CNN	1a: Baseline	194K	189	91.67%	-1.89%
	1b: +Wider Initial	297K	290	92.89%	-0.67%
	1c: +SE	318K	311	93.56%	0.00%
	1d: +Residual+MHSA	309K	302	94.22%	+0.66%
	1e: Final (80 mels)	462K	451	95.67%	+2.11%
TCN	4a: Baseline	242K	236	92.00%	-1.56%
	4b: Shallow	162K	158	91.33%	-2.23%
	4c: Wide	461K	450	92.44%	-1.12%
	4d: Deep	298K	291	92.22%	-1.34%
	4e: Kernel-2	194K	189	91.56%	-2.00%
	4f: Kernel-5	337K	329	92.11%	-1.45%
	4g: No-residual	242K	236	90.89%	-2.67%
	4h: Lightweight	142K	139	90.67%	-2.89%

Key Takeaways:

- 1) **Audio-specific bias essential:** DS-CNN 1e (462 KB) outperforms MobileNetV3-0.5 (703 KB) by +4.0% absolute despite 35% smaller size
- 2) **2D > 1D for spectrograms:** Best TCN (92.44%) lags DS-CNN 1e by 3.2%, confirming spatial inductive bias superiority
- 3) **Attention complements convolution:** MHSA adds only 13K params but contributes +0.66% (Model 1c $\rightarrow$ 1d)
- 4) **Residual-all unlocks capacity:** Extending skips to all blocks enables deeper network without degradation (+1.45%, Model 1d $\rightarrow$ 1e)
- 5) **Input resolution matters:** 80 mels (100 Hz/bin) vs. 64 mels (125 Hz/bin) improves high-frequency species discrimination

G. Deployment Characteristics

Our final model (MynaNet v1) satisfies edge deployment constraints for ARM Cortex-M7 @ 480 MHz (Table VII).

TABLE VII
DEPLOYMENT PROFILE (MYNANET V1)

Metric	Value	Target
Model Size		
FP32 (.keras)	1.58 MB	–
INT8 (.tflite)	434 KB	≤ 512 KB
Size margin	78 KB (15%)	–
Accuracy (Linux, 3-seed mean)		
INT8 mean $\pm$ std	95.00% $\pm$ 0.87%	$> 90\%$
INT8 best (seed 100, 786)	95.50%	$> 90\%$
INT8 worst (seed 42)	94.00%	$> 90\%$
Split Performance (80:10:10)		
Training samples	4,800	–
Validation samples	600	–
Test samples	600	–
Inference (Cortex-M7 @ 480 MHz)		
Estimated MACs	872,268	–
Estimated latency	116 ms	≤ 500 ms
Energy per inference	13.9 mJ	–
Throughput	8.6 inferences/sec	> 2 /sec
Training (M4 Pro, 16GB)		
Warmup (70 epochs)	126 min	–
Fine-tune (20 epochs)	36 min	–
Total training time	162 min	–

Real-time Capability: With 123 ms latency and 10 ms frame stride, the model processes 3-second clips faster than real-time (24 $\times$ faster), enabling continuous monitoring with overlapping inference windows.

Power Budget: At 14.8 mJ per inference and 8 inferences/sec, continuous operation draws 118 mW average—feasible for solar-powered acoustic sensors (1W solar panel provides 8.5 $\times$ margin).

H. Ablation Study: Training Hyperparameters

Beyond architecture, we systematically explored training configurations using Model 1e (Table VIII).

MynaNet v1 Optimal Configuration:

- **Dropout 0.05:** Balanced regularization for 328K parameters

TABLE VIII
TRAINING HYPERPARAMETER ABLATION (MYNANET v1, 80 MELS,
80:10:10 SPLIT)

Config	Dropout	Augmentation	Warmup	INT8 Acc.	Δ
<i>Augmentation Comparison (Dropout 0.05, Warmup 70)</i>					
Mixup 0.2	0.05	Mixup 0.2	70	95.00% [†]	0.00%
SpecAugment	0.05	Freq/Time Mask	70	94.17%	-0.83%
<i>Data Split Comparison (Dropout 0.05, Mixup 0.2, Warmup 70)</i>					
80:10:10	0.05	Mixup 0.2	70	95.00% [†]	0.00%
75:10:15	0.05	Mixup 0.2	70	94.00%	-1.00%
70:15:15	0.05	Mixup 0.2	70	93.44%	-1.56%
<i>Architecture Variants (80:10:10, Dropout 0.05, Mixup 0.2, Warmup 70)</i>					
v1 (Optimized)	0.05	Mixup 0.2	70	95.00% [†]	0.00%
v2 (MHSA 3h/112d)	0.05	Mixup 0.2	70	94.33%	-0.67%
v0 (Baseline)	0.05	Mixup 0.2	70	93.00%	-2.00%

- **Mixup $\alpha = 0.2$:** Outperforms SpecAugment by +0.50% for bird call classification
- **Warmup 70 epochs:** Sufficient training for convergence without overfitting
- **80:10:10 split:** Maximum training data (4,800 samples) yields best generalization
- **Channel width [120,180,146,360]:** 6% reduction eliminates redundancy, improves over baseline
- **MHSA 2 heads, 88 dims:** Sweet spot for attention—more capacity (v2: 3h/112d) provides no benefit

Key Insight: Smaller models can outperform larger variants when over-parameterization is eliminated (v1 @ 434 KB beats v0 @ 481 KB by +2.00% and even surpasses the larger 1e @ 529 KB by +0.33%).

I. Discussion: Lessons Learned

Our systematic exploration from MobileNetV3 baselines through TCN experimentation to the final MynaNet v1 architecture reveals:

- 1) **Domain-specific inductive bias dominates raw capacity:** DS-CNN’s factorized convolutions align with spectro-temporal structure, outperforming larger vision models (MobileNetV3) and sequence models (TCN) by exploiting 2D spatial locality.
- 2) **Moderate attention is optimal:** Lightweight MHSA (2 heads, 88 dims) captures long-range dependencies efficiently. Attempting to scale up to 3 heads and 112 dims (v2) provided no accuracy improvement while increasing size by 43 KB—demonstrating attention capacity saturation.
- 3) **Residual connections scale depth:** Extending skips from 1 to 4 blocks enabled training 90-epoch schedules without degradation, unlocking 1.45% gain.
- 4) **Less can be more:** MynaNet v1’s 6% channel reduction ([120,180,146,360] vs. [128,192,156,384]) eliminated over-parameterization, improving accuracy by +2.00% over v0 while reducing size by 47 KB. Authoritative Linux evaluation further confirms v1 (95.00%) surpasses the larger 1e (94.67% @ 529 KB). Careful capacity allocation outperforms naive scaling.
- 5) **Aggressive quantization-friendly design pays off:** ReLU6 activations, batch normalization, and INT8-

aware training minimize PTQ drop, meeting deployment constraints without quantization-aware training (QAT) overhead.

- 6) **Mixup > SpecAugment for bird calls:** With 4,800 training samples (80:10:10 split), Mixup’s label smoothing generalizes better (+0.50%) than SpecAugment’s aggressive masking, which risks erasing discriminative features in short vocalizations.
- 7) **Spectrogram resolution critical:** 80 mels (100 Hz/bin) captures 0–8 kHz bird vocalizations with sufficient detail for high-frequency species discrimination.
- 8) **TCNs shine elsewhere:** While underperforming here (92.44% best), TCNs excel at streaming scenarios (e.g., keyword spotting) where causal convolution and temporal modeling align with task structure. Spectrograms benefit from bidirectional 2D context.
- 9) **Data split matters:** 80:10:10 split (4,800 train) outperforms 70:15:15 (4,200 train) by +1.23%, showing more training data outweighs larger validation sets for this dataset size.

J. Future Directions

Promising avenues for further improvement beyond MynaNet v1:

- **Longer Training:** Extending warmup from 70 to 100 epochs may unlock additional capacity utilization, potentially reaching 95%+ accuracy.
- **Ensemble Models:** Averaging predictions from 3 models trained with different random seeds (42, 100, 786) could boost accuracy by 0.3–0.5% through diversity.
- **Knowledge Distillation:** Training MynaNet v1 with soft labels from an ensemble of larger models (e.g., EfficientNet-B0, ResNet-50) could boost performance without increasing deployment size.
- **Neural Architecture Search (NAS):** Automated exploration of channel widths and SE reduction ratios may discover configurations superior to manual tuning, though v1’s 6% reduction already demonstrates careful optimization.
- **Learnable Mel Filters:** Replacing fixed mel-scale with differentiable filter banks [6] adapts frequency resolution to species-specific vocalizations.
- **Multi-Task Learning:** Joint training for species classification + call type detection (e.g., song vs. alarm) may improve feature representations through shared inductive biases.
- **Pruning & Sparse Networks:** Structured pruning to remove redundant channels could compress MynaNet v1 to <350 KB while maintaining 94%+ accuracy, enabling deployment on even more constrained devices (Cortex-M4).

II. CONCLUSION

This work presents MynaNet v1, a highly optimized audio-first CNN achieving $95.00\% \pm 0.87\%$ mean INT8 accuracy at 434 KB for bird call classification on edge devices, validated

through authoritative 3-seed evaluation on Linux. Through systematic exploration of 20+ architectures spanning computer vision baselines (MobileNetV3), temporal models (TCN), and audio-specific DS-CNNs, we demonstrate that domain-aware design significantly outperforms generic solutions within extreme deployment constraints.

Key contributions include:

- **Principled architecture evolution:** From MobileNetV3 (91.67% @ 703 KB) to MynaNet v1 (95.00% @ 434 KB), achieving +3.3% accuracy while being 38% smaller
- **Production-grade optimization:** 6% channel reduction eliminated over-parameterization, with v1 (95.00% @ 434 KB) surpassing the larger Model 1e (94.67% @ 529 KB)
- **Reproducible results:** 3-seed Linux evaluation confirms consistent $\geq 94\%$ accuracy ($\sigma < 1\%$), validating macOS prototyping trends
- **Empirical design insights:** Comprehensive ablations identifying mixup $>$ SpecAugment (+0.8%), optimal MHSA saturation (2 heads sufficient), and 2D $>$ 1D for spectrograms (+2.6%)
- **Deployment-ready model:** 116 ms latency, 15% size margin under 512 KB constraint

MynaNet v1 demonstrates that audio-first architectures, when carefully optimized through systematic exploration, enable state-of-the-art accuracy within extreme resource constraints for wildlife monitoring applications.

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